

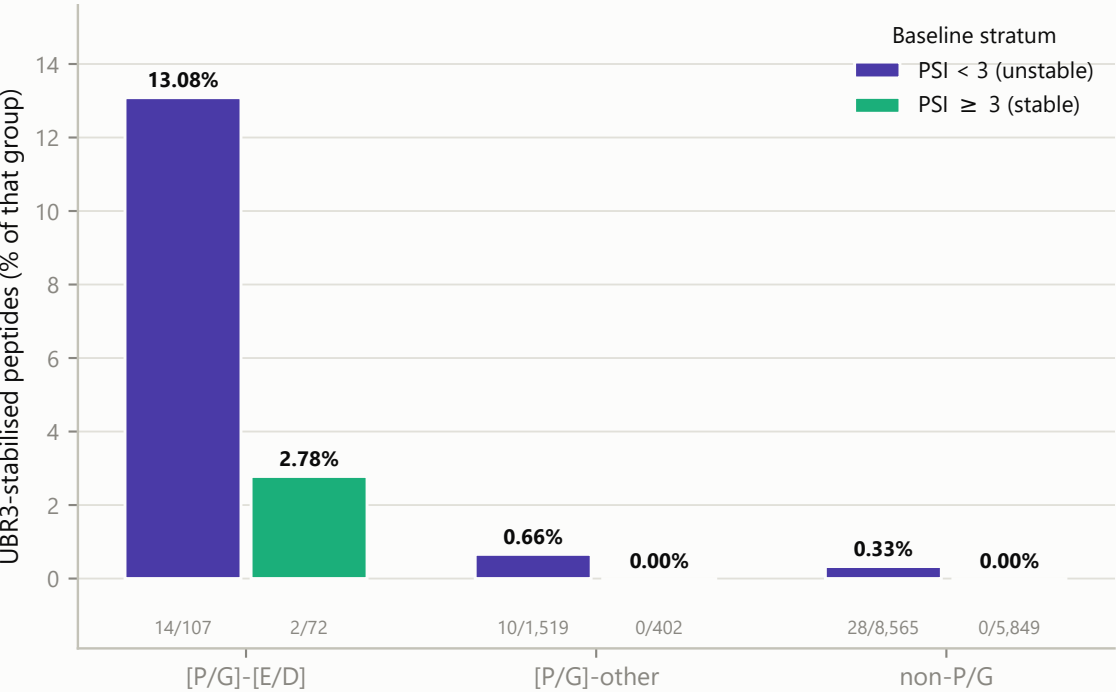
Figure 7 | The [P/G]-[E/D] effect is not a by-product of baseline instability

Substrates start unstable, so the motif could in principle be enriched simply because Pro/Gly N-termini are unstable. Stratifying by control PSI shows it is not: the motif is evenly distributed across strata yet enriches for substrates within each one.

A

The motif effect survives stratification

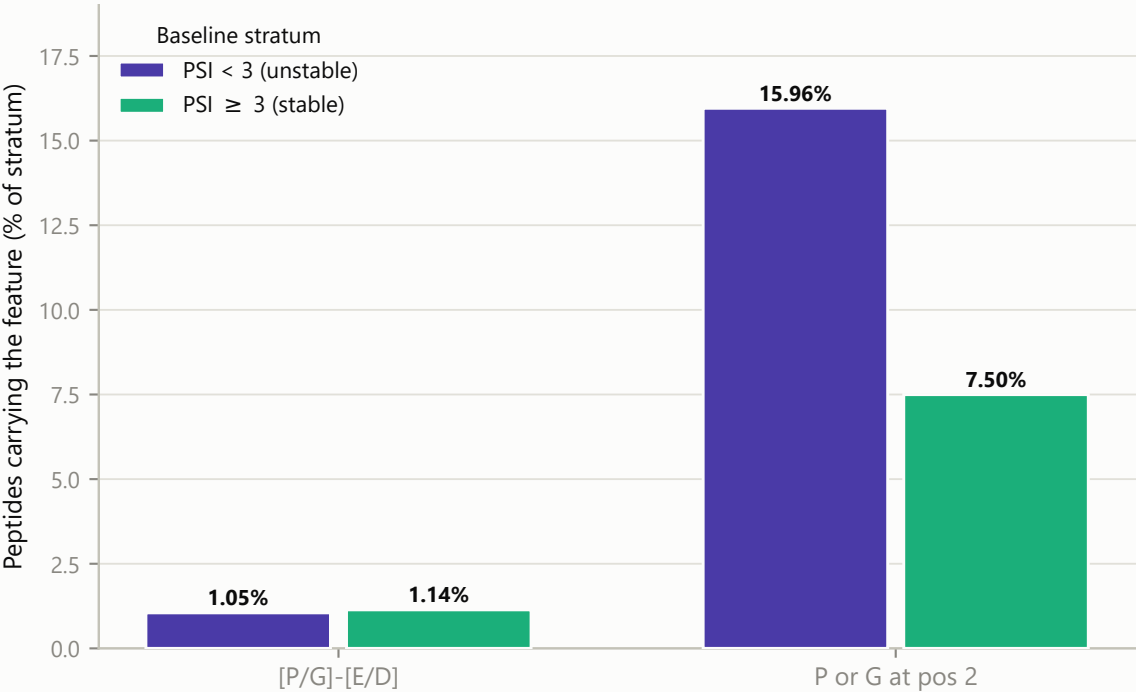
Within unstable peptides alone: 13.1% vs 0.33%, Fisher OR 45.9, $p = 3 \times 10^{-17}$



B

[P/G]-[E/D] is not over-represented among unstable peptides

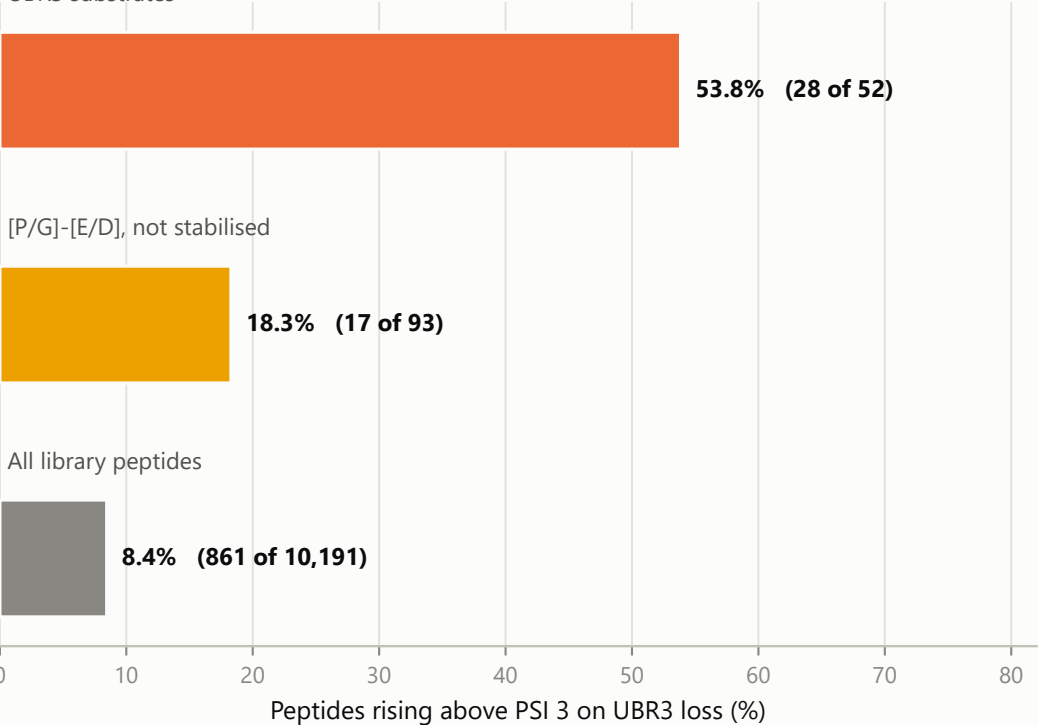
Bare Pro/Gly is skewed (16.0% vs 7.5%), so the acidic residue carries the UBR3 dependence



C

Crossing is graded, not all-or-nothing

Peptides starting below PSI 3; non-substrates with the motif still cross at twice background UBR3 substrates



D

Within unstable peptides, the motif still shifts ΔPSI

Only peptides starting below PSI 3; boxes are median and quartiles

